

The inverse of $X_{-1}(u)$ in the group $G(\mathfrak{m})$: a machine-checked proof

Abstract

We present a formal verification of Corollary 3.2 (label `cor: gsimple x inverse`) from a text on the Monster Lie algebra group $G(\mathfrak{m})$. The group is defined by generators and relations, and the corollary states that $X_{-1}(u)^{-1} = X_{-1}(-u)$ for every $u \in \mathbb{C}$. In the formalization the group is the presented group on the generating symbols $\mathcal{X}$ modulo the full list of defining relations. The index set E is built from an arbitrary function $c : \mathbb{N} \rightarrow \mathbb{N}$ in place of the coefficients of the modular function J . The identity is proved in Lean 4 with Mathlib for every such c , using only the relation (Re:1a).

Source	Carbone et al., G-Simple (draft), GSimple_Corollary_3_2
Lean	leanprover/lean4:v4.23.0, Mathlib 37df177aaa
Machine checks	compiles; no <code>sorry</code> ; statement matches target; module build
Axioms	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
Independent review	FAITHFUL (claude-opus-5-5)
Agents	captain claude-sonnet-5-5; solvers aiprover; report claude-opus-5-5
Run	srv_20261004_123234_GSimple_Corollary_3_2

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1 Statement

Let $\mathbb{C}$ denote the complex numbers and $\mathbb{N}$ the natural numbers. Let $c(j)$ denote the coefficient of q^j in the modular function

$$J(q) = \sum_{j \geq -1} c(j) q^j = \frac{1}{q} + 196884 q + 21493760 q^2 + 864299970 q^3 + \dots,$$

so that $c(-1) = 1$, $c(0) = 0$, $c(1) = 196884, \dots$

Index set and constants. The index set is

$$E = \{(\ell, j, k) \mid j \in \mathbb{N}, 1 \leq k \leq c(j), 0 \leq \ell < j\}.$$

For $(\ell, j, k) \in E$ we set

$$c_{\ell j} := (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell) \in \mathbb{C}.$$

Generators. Let $\mathcal{X}$ be the set of symbols

$$\{H_1(s), H_2(s), X_{-1}(u), Y_{-1}(u), X_{\ell,jk}(u), Y_{\ell,jk}(u) \mid s \in \mathbb{C}^\times, u \in \mathbb{C}, (\ell, j, k) \in E\},$$

and let $F(\mathcal{X})$ be the free group on $\mathcal{X}$.

Relations. The commutator is $(a, b) := aba^{-1}b^{-1}$. The set $\mathcal{R} \subseteq F(\mathcal{X})$ and the auxiliary elements $\tilde{w}_{-1}(s)$, $\tilde{w}_{\ell,jk}(s)$ are given as follows, for all $s, t \in \mathbb{C}^\times$, $u, v \in \mathbb{C}$ and $(\ell, j, k), (m, p, q) \in E$.

$$\tilde{w}_{-1}(s) := X_{-1}(s) Y_{-1}(-s^{-1}) X_{-1}(s), \quad \tilde{w}_{-1} := \tilde{w}_{-1}(1), \quad (\text{Re:0})$$

$$\tilde{w}_{\ell,jk}(s) := X_{\ell,jk}(s) Y_{\ell,jk}\left(\frac{-s^{-1}}{c_{\ell j}}\right) X_{\ell,jk}(s), \quad \tilde{w}_{\ell,jk} := \tilde{w}_{\ell,jk}(1), \quad (\text{Im:0})$$

Relations defining a subgroup $\text{GL}_2(-1)$:

$$H_1(s)H_1(t) = H_1(st), \quad (\text{H:1a})$$

$$H_2(s)H_2(t) = H_2(st), \quad (\text{H:1b})$$

$$H_1(s)H_2(t) = H_2(t)H_1(s), \quad (\text{H:2})$$

$$X_{-1}(u)X_{-1}(v) = X_{-1}(u+v), \quad (\text{Re:1a})$$

$$Y_{-1}(u)Y_{-1}(v) = Y_{-1}(u+v), \quad (\text{Re:1b})$$

$$Y_{-1}(-t)X_{-1}(s)Y_{-1}(t) = X_{-1}(-t^{-1})Y_{-1}(-t^2s)X_{-1}(t^{-1}), \quad (\text{Re:2})$$

$$\tilde{w}_{-1}(s)\tilde{w}_{-1} = H_1(-s)H_2(-s^{-1}), \quad (\text{Re:3})$$

$$\tilde{w}_{-1}X_{-1}(u)\tilde{w}_{-1}^{-1} = Y_{-1}(-u), \quad (\text{Re:4a})$$

$$\tilde{w}_{-1}Y_{-1}(u)\tilde{w}_{-1}^{-1} = X_{-1}(-u), \quad (\text{Re:4b})$$

$$\tilde{w}_{-1}H_1(s)\tilde{w}_{-1}^{-1} = H_2(s), \quad (\text{Re:5a})$$

$$\tilde{w}_{-1}H_2(s)\tilde{w}_{-1}^{-1} = H_1(s), \quad (\text{Re:5b})$$

$$H_1(s)X_{-1}(u)H_1(s)^{-1} = X_{-1}(su), \quad (\text{Re:6a})$$

$$H_2(s)X_{-1}(u)H_2(s)^{-1} = X_{-1}(s^{-1}u), \quad (\text{Re:6b})$$

$$H_1(s)Y_{-1}(u)H_1(s)^{-1} = Y_{-1}(s^{-1}u), \quad (\text{Re:6c})$$

$$H_2(s)Y_{-1}(u)H_2(s)^{-1} = Y_{-1}(su), \quad (\text{Re:6d})$$

Relations defining subgroups $\text{GL}_2(\ell, j, k)$:

$$X_{\ell,jk}(u)X_{\ell,jk}(v) = X_{\ell,jk}(u+v), \quad (\text{Im:1a})$$

$$Y_{\ell,jk}(u)Y_{\ell,jk}(v) = Y_{\ell,jk}(u+v), \quad (\text{Im:1b})$$

$$Y_{\ell,jk}(-t)X_{\ell,jk}(s)Y_{\ell,jk}(t) = X_{\ell,jk}\left(\frac{-t^{-1}}{c_{\ell j}}\right)Y_{\ell,jk}(-c_{\ell j}t^2s)X_{\ell,jk}\left(\frac{t^{-1}}{c_{\ell j}}\right), \quad (\text{Im:2})$$

$$\tilde{w}_{\ell,jk}(s^{(\ell+1)(j-\ell)})\tilde{w}_{\ell,jk} = H_1((-s)^{j-\ell})H_2((-s)^{\ell+1}), \quad (\text{Im:3})$$

$$\tilde{w}_{\ell,jk}X_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = Y_{\ell,jk}\left(\frac{-u}{c_{\ell j}}\right), \quad (\text{Im:4a})$$

$$\tilde{w}_{\ell,jk}Y_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = X_{\ell,jk}(-c_{\ell j}u), \quad (\text{Im:4b})$$

$$\tilde{w}_{\ell,jk}H_1(s^{j-\ell})\tilde{w}_{\ell,jk}^{-1} = H_2(s^{-(\ell+1)}), \quad (\text{Im:5a})$$

$$\tilde{w}_{\ell,jk}H_2(s^{\ell+1})\tilde{w}_{\ell,jk}^{-1} = H_1(s^{-(j-\ell)}), \quad (\text{Im:5b})$$

$$H_1(s)X_{\ell,jk}(u)H_1(s)^{-1} = X_{\ell,jk}(s^{\ell+1}u), \quad (\text{Im:6a})$$

$$H_2(s)X_{\ell,jk}(u)H_2(s)^{-1} = X_{\ell,jk}(s^{j-\ell}u), \quad (\text{Im:6b})$$

$$H_1(s)Y_{\ell,jk}(u)H_1(s)^{-1} = Y_{\ell,jk}(s^{-(\ell+1)}u), \quad (\text{Im:6c})$$

$$H_2(s)Y_{\ell,jk}(u)H_2(s)^{-1} = Y_{\ell,jk}(s^{-(j-\ell)}u), \quad (\text{Im:6d})$$

Relations between the unipotent generators:

$$(X_{-1}(u), X_{j-1,jk}(v)) = 1, \quad (\text{U:1a})$$

$$(Y_{-1}(u), Y_{j-1,jk}(v)) = 1, \quad (\text{U:1b})$$

$$(Y_{-1}(u), X_{0,jk}(v)) = 1, \quad (\text{U:1c})$$

$$(X_{-1}(u), Y_{0,jk}(v)) = 1, \quad (\text{U:1d})$$

$$(X_{\ell,jk}(u), Y_{m,pq}(v)) = 1 \quad \text{for } j \neq p, \text{ or } k \neq q, \text{ or } |\ell - m| > 1, \quad (\text{U:2})$$

$$(X_{1,2k}(u), Y_{0,2k}(v)) = X_{-1}(uv), \quad (\text{U:3a})$$

$$(X_{0,2k}(u), Y_{1,2k}(v)) = Y_{-1}(uv), \quad (\text{U:3b})$$

Action of $\tilde{w}_{-1}$ on the unipotent generators:

$$\tilde{w}_{-1}X_{\ell,jk}(u)\tilde{w}_{-1}^{-1} = X_{j-1-\ell,jk}((-1)^{j-\ell-1}u), \quad (\text{U:4a})$$

$$\tilde{w}_{-1}Y_{\ell,jk}(u)\tilde{w}_{-1}^{-1} = Y_{j-1-\ell,jk}((-1)^{j-\ell-1}u). \quad (\text{U:4b})$$

The group. The group is

$$G(\mathfrak{m}) = \langle \mathcal{X} \mid \mathcal{R} \rangle = F(\mathcal{X})/N_{\mathcal{R}},$$

where $N_{\mathcal{R}}$ is the normal closure of $\mathcal{R}$.

Theorem 1.1 (Corollary 3.2). *For all $u \in \mathbb{C}$, $X_{-1}(u)^{-1} = X_{-1}(-u)$ in $G(\mathfrak{m})$.*

2 Formalization

Throughout, $c : \mathbb{N} \rightarrow \mathbb{N}$ is an arbitrary function. It is a parameter of every definition, and it replaces the coefficient function of J ; see the discussion after the target statement.

The index set. The index set E_c is the set of triples $(\ell, j, k) \in \mathbb{N}^3$ with

$$\ell < j \quad \text{and} \quad 1 \leq k \leq c(j).$$

This is the source's E with c in place of the coefficients of J .

Lean: Idx

```

| /-- Index set  $E = \{(\ell, j, k) \mid 1 \leq k \leq c(j), 0 \leq \ell < j\}$ , encoded as triples  $(\ell, j, k)$ . -/
| def Idx (c :  $\mathbb{N} \rightarrow \mathbb{N}$ ) : Type := {t :  $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$  // t.1 < t.2.1  $\wedge$  1  $\leq$  t.2.2  $\wedge$  t.2.2  $\leq$  c t.2.1}

```

For $e = (\ell, j, k) \in E_c$ we write $\ell_e = \ell$, $j_e = j$ and $k_e = k$ for the three coordinates.

Lean: Idx.l

```

| def Idx.l (e : Idx c) :  $\mathbb{N}$  := e.1.1

```

Lean: Idx.j

```

| def Idx.j (e : Idx c) :  $\mathbb{N}$  := e.1.2.1

```

Lean: Idx.k

```

| def Idx.k (e : Idx c) :  $\mathbb{N}$  := e.1.2.2

```

The generating symbols. The set of symbols $\mathcal{X}_c$ is the disjoint union of the following families:

- $H_1(s)$ and $H_2(s)$ for $s \in \mathbb{C}^\times$;
- $X_{-1}(u)$ and $Y_{-1}(u)$ for $u \in \mathbb{C}$;
- $X_e(u)$ and $Y_e(u)$ for $e \in E_c$ and $u \in \mathbb{C}$.

Here $X_e(u)$ stands for the source's $X_{\ell, j, k}(u)$ when $e = (\ell, j, k)$, and similarly for $Y_e(u)$.

Lean: Gen

```

| /-- The generating symbols  $\mathcal{X}$ . -/
| inductive Gen (c :  $\mathbb{N} \rightarrow \mathbb{N}$ ) : Type
| H1 (s :  $\mathbb{C}^\times$ ) : Gen c
| H2 (s :  $\mathbb{C}^\times$ ) : Gen c
| Xm (u :  $\mathbb{C}$ ) : Gen c
| Ym (u :  $\mathbb{C}$ ) : Gen c
| X (e : Idx c) (u :  $\mathbb{C}$ ) : Gen c
| Y (e : Idx c) (u :  $\mathbb{C}$ ) : Gen c

```

The next six definitions are the images of these symbols in the free group $F(\mathcal{X}_c)$, that is, the corresponding words of length one. We denote them by the same letters $H_1(s)$, $H_2(s)$, $X_{-1}(u)$, $Y_{-1}(u)$, $X_e(u)$, $Y_e(u)$.

Lean: fH1

```

| def fH1 (s :  $\mathbb{C}^\times$ ) : FreeGroup (Gen c) := FreeGroup.of (Gen.H1 s)

```

Lean: fH2

```
| def fH2 (s : ℂ×) : FreeGroup (Gen c) := FreeGroup.of (Gen.H2 s)
```

Lean: fXm

```
| def fXm (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Xm u)
```

Lean: fYm

```
| def fYm (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Ym u)
```

Lean: fX

```
| def fX (e : Idx c) (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.X e u)
```

Lean: fY

```
| def fY (e : Idx c) (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Y e u)
```

The constants $c_{\ell j}$. For $e = (\ell, j, k) \in E_c$,

$$c_e := (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell) \in \mathbb{C}.$$

The differences $j-1$ and $j-\ell$ are computed in $\mathbb{N}$ by truncated subtraction. Since $\ell < j$ forces $j \geq 1$, both are the true differences. Hence $c_e = c_{\ell j}$.

Lean: cc

```
| /-- c_{\ell j} = (-1)^{\ell+1} C(j-1, \ell) (\ell+1)(j-\ell). -/
| def cc (e : Idx c) : ℂ :=
|   ((-1 : ℂ) ^ (e.1 + 1)) * (((e.j - 1).choose e.1 : ℕ) : ℂ) * ((e.1 : ℂ) + 1) * ((e.j - e.1 : ℕ) : ℂ)
```

The elements $\tilde{w}$. For $s \in \mathbb{C}^\times$ and $e \in E_c$, define the words in $F(\mathcal{X}_c)$

$$\tilde{w}_{-1}(s) := X_{-1}(s) Y_{-1}(-s^{-1}) X_{-1}(s)$$

and

$$\tilde{w}_e(s) := X_e(s) Y_e(-s^{-1}/c_e) X_e(s).$$

These are (Re:0) and (Im:0). The source's $\tilde{w}_{-1}$ and $\tilde{w}_{\ell,j,k}$ are $\tilde{w}_{-1}(1)$ and $\tilde{w}_e(1)$.

Lean: wm

```

/-- (Re:0) -/
def wm (s :  $\mathbb{C}^\times$ ) : FreeGroup (Gen c) := fXm (s :  $\mathbb{C}$ ) * fYm ( $-(s : \mathbb{C})^{-1}$ ) * fXm (s :  $\mathbb{C}$ )

```

Lean: wt

```

/-- (Im:0) -/
def wt (e : Idx c) (s :  $\mathbb{C}^\times$ ) : FreeGroup (Gen c) :=
  fX e (s :  $\mathbb{C}$ ) * fY e ( $-(s : \mathbb{C})^{-1}$ ) / cc e * fX e (s :  $\mathbb{C}$ )

```

Commutator and conjugation. For $a, b \in F(\mathcal{X}_c)$, the commutator is $(a, b) := aba^{-1}b^{-1}$ and the conjugate is $a b a^{-1}$.

Lean: comm

```

/-- commutator (a,b) = a b a^{-1} b^{-1} -/
def comm (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a^{-1} * b^{-1}

```

Lean: conj

```

/-- conjugation a b a^{-1} -/
def conj (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a^{-1}

```

The relators. The set $\mathcal{R}_c \subseteq F(\mathcal{X}_c)$ consists of the words LR^{-1} , one for each source relation $L = R$. For relations of the form $(a, b) = 1$ the word is the commutator (a, b) itself. The families are listed below. Here s, t range over $\mathbb{C}^\times$, u, v over $\mathbb{C}$, and e, f, e' over E_c . Powers of s with exponent in $\mathbb{N}$ or $\mathbb{Z}$ are taken in $\mathbb{C}^\times$ when they are arguments of H_i or of $\tilde{w}_e$, and in $\mathbb{C}$ when they are coefficients inside the arguments of X_e and Y_e . Abbreviate $\ell = \ell_e$, $j = j_e$, $k = k_e$.

- (H:1a), (H:1b): $H_1(s)H_1(t)H_1(st)^{-1}$ and $H_2(s)H_2(t)H_2(st)^{-1}$.
- (H:2): $H_1(s)H_2(t)(H_2(t)H_1(s))^{-1}$.
- (Re:1a), (Re:1b): $X_{-1}(u)X_{-1}(v)X_{-1}(u+v)^{-1}$ and $Y_{-1}(u)Y_{-1}(v)Y_{-1}(u+v)^{-1}$.
- (Re:2): $Y_{-1}(-t)X_{-1}(s)Y_{-1}(t)(X_{-1}(-t^{-1})Y_{-1}(-t^2s)X_{-1}(t^{-1}))^{-1}$.
- (Re:3): $\tilde{w}_{-1}(s)\tilde{w}_{-1}(1)(H_1(-s)H_2(-s^{-1}))^{-1}$.
- (Re:4a)–(Re:5b): $\tilde{w}_{-1}(1)g\tilde{w}_{-1}(1)^{-1}h^{-1}$ for the pairs (g, h) equal to $(X_{-1}(u), Y_{-1}(-u))$, $(Y_{-1}(u), X_{-1}(-u))$, $(H_1(s), H_2(s))$ and $(H_2(s), H_1(s))$.
- (Re:6a)–(Re:6d): $H_1(s)X_{-1}(u)H_1(s)^{-1}X_{-1}(su)^{-1}$, together with the analogous words for $X_{-1}(s^{-1}u)$ under H_2 , for $Y_{-1}(s^{-1}u)$ under H_1 , and for $Y_{-1}(su)$ under H_2 .
- (Im:1a), (Im:1b): $X_e(u)X_e(v)X_e(u+v)^{-1}$ and $Y_e(u)Y_e(v)Y_e(u+v)^{-1}$.
- (Im:2): $Y_e(-t)X_e(s)Y_e(t)(X_e(-t^{-1}/c_e)Y_e(-c_et^2s)X_e(t^{-1}/c_e))^{-1}$.

- (Im:3): $\tilde{w}_e(s^{(\ell+1)(j-\ell)})\tilde{w}_e(1)(H_1((-s)^{j-\ell})H_2((-s)^{\ell+1}))^{-1}$.
- (Im:4a), (Im:4b): $\tilde{w}_e(1)X_e(u)\tilde{w}_e(1)^{-1}Y_e(-u/c_e)^{-1}$ and $\tilde{w}_e(1)Y_e(u)\tilde{w}_e(1)^{-1}X_e(-c_e u)^{-1}$.
- (Im:5a), (Im:5b): $\tilde{w}_e(1)H_1(s^{j-\ell})\tilde{w}_e(1)^{-1}H_2(s^{-(\ell+1)})^{-1}$ and $\tilde{w}_e(1)H_2(s^{\ell+1})\tilde{w}_e(1)^{-1}H_1(s^{-(j-\ell)})^{-1}$. The negative exponents are integer powers in $\mathbb{C}^\times$.
- (Im:6a)–(Im:6d): $H_1(s)X_e(u)H_1(s)^{-1}X_e(s^{\ell+1}u)^{-1}$, together with the analogous words for $X_e(s^{j-\ell}u)$ under H_2 , for $Y_e(s^{-(\ell+1)}u)$ under H_1 , and for $Y_e(s^{-(j-\ell)}u)$ under H_2 .
- (U:1a), (U:1b): $(X_{-1}(u), X_e(v))$ and $(Y_{-1}(u), Y_e(v))$ for all e with $\ell_e + 1 = j_e$, that is, $\ell_e = j_e - 1$.
- (U:1c), (U:1d): $(Y_{-1}(u), X_e(v))$ and $(X_{-1}(u), Y_e(v))$ for all e with $\ell_e = 0$.
- (U:2): $(X_e(u), Y_f(v))$ for all e, f with $j_e \neq j_f$, or $k_e \neq k_f$, or $|\ell_e - \ell_f| > 1$.
- (U:3a): $(X_e(u), Y_f(v)) X_{-1}(uv)^{-1}$ for all $e, f \in E_c$ such that $e = (1, 2, k)$ and $f = (0, 2, k)$ for some $k \in \mathbb{N}$.
- (U:3b): $(X_e(u), Y_f(v)) Y_{-1}(uv)^{-1}$ for all $e, f \in E_c$ such that $e = (0, 2, k)$ and $f = (1, 2, k)$ for some k .
- (U:4a), (U:4b): $\tilde{w}_{-1}(1)X_e(u)\tilde{w}_{-1}(1)^{-1}X_{e'}((-1)^{j-\ell-1}u)^{-1}$ and the same word with Y in place of X , for all e, e' with $e' = (j - 1 - \ell, j, k)$. Because $\ell < j$, the subtraction $j - 1 - \ell$ in $\mathbb{N}$ is exact, and this triple always lies in E_c .

This list is exactly the source's $\mathcal{R}$, with E replaced by E_c .

Lean: R1

```

/-- The set of defining relators; each constructor says `lhs = rhs` via the word `lhs * rhs⁻¹`. -/
inductive R1 (c : ℕ → ℕ) : FreeGroup (Gen c) → Prop
| h1a (s t : ℂ×) : R1 c (fH1 s * fH1 t * (fH1 (s * t))-1)
| h1b (s t : ℂ×) : R1 c (fH2 s * fH2 t * (fH2 (s * t))-1)
| h2 (s t : ℂ×) : R1 c (fH1 s * fH2 t * (fH2 t * fH1 s)-1)
| re1a (u v : ℂ) : R1 c (fXm u * fXm v * (fXm (u + v))-1)
| re1b (u v : ℂ) : R1 c (fYm u * fYm v * (fYm (u + v))-1)
| re2 (s t : ℂ×) :
  R1 c (fYm (-(t : ℂ)) * fXm (s : ℂ) * fYm (t : ℂ) *
    (fXm (-(t : ℂ)-1) * fYm (-(t : ℂ) ^ 2 * (s : ℂ)) * fXm ((t : ℂ)-1)-1)
| re3 (s : ℂ×) : R1 c (wm s * wm 1 * (fH1 (-s) * fH2 (-s-1))-1)
| re4a (u : ℂ) : R1 c (conj (wm 1) (fXm u) * (fYm (-u))-1)
| re4b (u : ℂ) : R1 c (conj (wm 1) (fYm u) * (fXm (-u))-1)
| re5a (s : ℂ×) : R1 c (conj (wm 1) (fH1 s) * (fH2 s)-1)
| re5b (s : ℂ×) : R1 c (conj (wm 1) (fH2 s) * (fH1 s)-1)
| re6a (s : ℂ×) (u : ℂ) : R1 c (conj (fH1 s) (fXm u) * (fXm ((s : ℂ) * u))-1)
| re6b (s : ℂ×) (u : ℂ) : R1 c (conj (fH2 s) (fXm u) * (fXm ((s : ℂ)-1 * u))-1)
| re6c (s : ℂ×) (u : ℂ) : R1 c (conj (fH1 s) (fYm u) * (fYm ((s : ℂ)-1 * u))-1)
| re6d (s : ℂ×) (u : ℂ) : R1 c (conj (fH2 s) (fYm u) * (fYm ((s : ℂ) * u))-1)
| im1a (e : Idx c) (u v : ℂ) : R1 c (fX e u * fX e v * (fX e (u + v))-1)
| im1b (e : Idx c) (u v : ℂ) : R1 c (fY e u * fY e v * (fY e (u + v))-1)
| im2 (e : Idx c) (s t : ℂ×) :
  R1 c (fY e (-(t : ℂ)) * fX e (s : ℂ) * fY e (t : ℂ) *
    (fX e (-(t : ℂ)-1) / cc e) * fY e (-(cc e) * (t : ℂ) ^ 2 * (s : ℂ)) *
    fX e ((t : ℂ)-1 / cc e)-1)
| im3 (e : Idx c) (s : ℂ×) :
  R1 c (wt e (s ^ ((e.1 + 1) * (e.j - e.1))) * wt e 1 *

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      (fH1 ((-s) ^ (e.j - e.l)) * fH2 ((-s) ^ (e.l + 1)))-1
| im4a (e : Idx c) (u : ℂ) : R1 c (conj (wt e 1) (fX e u) * (fY e (-u / cc e))-1)
| im4b (e : Idx c) (u : ℂ) : R1 c (conj (wt e 1) (fY e u) * (fX e (-cc e * u))-1)
| im5a (e : Idx c) (s : ℂ×) :
  R1 c (conj (wt e 1) (fH1 (s ^ (e.j - e.l))) * (fH2 (s ^ (-((e.l : ℤ) + 1))))-1)
| im5b (e : Idx c) (s : ℂ×) :
  R1 c (conj (wt e 1) (fH2 (s ^ (e.l + 1))) * (fH1 (s ^ (-((e.j : ℤ) - e.l))))-1)
| im6a (e : Idx c) (s : ℂ×) (u : ℂ) :
  R1 c (conj (fH1 s) (fX e u) * (fX e ((s : ℂ) ^ (e.l + 1) * u))-1)
| im6b (e : Idx c) (s : ℂ×) (u : ℂ) :
  R1 c (conj (fH2 s) (fX e u) * (fX e ((s : ℂ) ^ (e.j - e.l) * u))-1)
| im6c (e : Idx c) (s : ℂ×) (u : ℂ) :
  R1 c (conj (fH1 s) (fY e u) * (fY e ((s : ℂ) ^ (-((e.l : ℤ) + 1) * u))-1)
| im6d (e : Idx c) (s : ℂ×) (u : ℂ) :
  R1 c (conj (fH2 s) (fY e u) * (fY e ((s : ℂ) ^ (-((e.j : ℤ) - e.l) * u))-1)
| u1a (e : Idx c) (h : e.l + 1 = e.j) (u v : ℂ) : R1 c (comm (fXm u) (fX e v))
| u1b (e : Idx c) (h : e.l + 1 = e.j) (u v : ℂ) : R1 c (comm (fYm u) (fY e v))
| u1c (e : Idx c) (h : e.l = 0) (u v : ℂ) : R1 c (comm (fYm u) (fX e v))
| u1d (e : Idx c) (h : e.l = 0) (u v : ℂ) : R1 c (comm (fXm u) (fY e v))
| u2 (e f : Idx c) (h : e.j ≠ f.j ∨ e.k ≠ f.k ∨ 1 < ((e.l : ℤ) - f.l)).natAbs (u v : ℂ) :
  R1 c (comm (fX e u) (fY f v))
| u3a (e f : Idx c) (k : ℕ) (he : e.1 = (1, 2, k)) (hf : f.1 = (0, 2, k)) (u v : ℂ) :
  R1 c (comm (fX e u) (fY f v) * (fXm (u * v))-1)
| u3b (e f : Idx c) (k : ℕ) (he : e.1 = (0, 2, k)) (hf : f.1 = (1, 2, k)) (u v : ℂ) :
  R1 c (comm (fX e u) (fY f v) * (fYm (u * v))-1)
| u4a (e e' : Idx c) (h : e'.1 = (e.j - 1 - e.l, e.j, e.k)) (u : ℂ) :
  R1 c (conj (wm 1) (fX e u) * (fX e' ((-1 : ℂ) ^ (e.j - e.l - 1) * u))-1)
| u4b (e e' : Idx c) (h : e'.1 = (e.j - 1 - e.l, e.j, e.k)) (u : ℂ) :
  R1 c (conj (wm 1) (fY e u) * (fY e' ((-1 : ℂ) ^ (e.j - e.l - 1) * u))-1)

```

The group. The group G_c is the presented group $\langle \mathcal{X}_c \mid \mathcal{R}_c \rangle = F(\mathcal{X}_c)/N_{\mathcal{R}_c}$, where $N_{\mathcal{R}_c}$ is the normal closure of $\mathcal{R}_c$. We write $\pi : F(\mathcal{X}_c) \rightarrow G_c$ for the quotient map.

Lean: G

```

/-- The group G(m) = <X | R>. -/
abbrev G (c : ℕ → ℕ) : Type := PresentedGroup {r : FreeGroup (Gen c) | R1 c r}

```

The target statement. Lean: target statement `gsimple_x_inverse`, fixed before the proof; solution is checked to have exactly this type

```

theorem gsimple_x_inverse (c : ℕ → ℕ) (u : ℂ) :
  ((PresentedGroup.of (GM.Gen.Xm (c := c) u) : GM.G c))-1 =
  (PresentedGroup.of (GM.Gen.Xm (c := c) (-u)) : GM.G c)

```

The statement reads: for every function $c : \mathbb{N} \rightarrow \mathbb{N}$ and every $u \in \mathbb{C}$,

$$\pi(X_{-1}(u))^{-1} = \pi(X_{-1}(-u)) \quad \text{in } G_c.$$

Relation to the source. The source's group $G(\mathbf{m})$ is the instance of G_c in which $c(j)$ is the coefficient of q^j in J for $j \in \mathbb{N}$.

- The values $c(-1)$ and $c(0)$ never enter E . For $c(-1)$ this is because $j \in \mathbb{N}$. For $c(0)$ it is because no ℓ satisfies $\ell < 0$.
- The formal theorem is stated for every c , with no hypothesis. In particular it holds for the coefficient function of J , and so it contains the source's corollary.
- The hypothesis $c(2) \geq 1$ mentioned in the problem set is not needed here.
- The corollary does not depend on c at all, since its proof uses only (Re:1a).

3 Proof

3.1 Overview

The source proof has two steps. It reads (Re:1a) as $X_{-1}(u)X_{-1}(v) = X_{-1}(u + v)$ in $G(\mathbf{m})$, and it sets $v = -u$ and uses $X_{-1}(0) = 1$, which it takes from Lemma 3.1. The formal proof follows the same two steps.

- Lemma 3.2 is the reading of (Re:1a) as an identity in G_c .
- Lemma 3.3 is the fact $X_{-1}(0) = 1$.

The only deviation is that Lemma 3.1 of the source is not imported. The fact $X_{-1}(0) = 1$ is instead derived directly from the instance $u = v = 0$ of (Re:1a). Lemma 3.1 records the general fact that every relator becomes trivial in G_c . It is not used by the later declarations, which invoke the same fact directly for the specific relators they need.

Lemma 3.1 (mk_rel_eq_one_in_G). *Let $c : \mathbb{N} \rightarrow \mathbb{N}$ and let $r \in F(\mathcal{X}_c)$ with $r \in \mathcal{R}_c$. Then $\pi(r) = 1$ in G_c .*

Lean: mk_rel_eq_one_in_G — proved by AIProver, job 1, session 0

```
theorem mk_rel_eq_one_in_G (c : ℕ → ℕ) (r : FreeGroup (GM.Gen c)) (h : GM.R1 c r) :
  (PresentedGroup.mk {r' : FreeGroup (GM.Gen c) | GM.R1 c r'} r : GM.G c) = 1 := by
  exact (PresentedGroup.one_of_mem h)
```

Proof. We have $r \in \mathcal{R}_c \subseteq N_{\mathcal{R}_c} = \ker \pi$. □

Lemma 3.2 (Xm_add_in_G). *Let $c : \mathbb{N} \rightarrow \mathbb{N}$ and $u, v \in \mathbb{C}$. Then in G_c*

$$\pi(X_{-1}(u)) \pi(X_{-1}(v)) = \pi(X_{-1}(u + v)).$$

Lean: Xm_add_in_G — proved by AIProver, job 2, session 1

```
theorem Xm_add_in_G (c : ℕ → ℕ) (u v : ℂ) :
  (PresentedGroup.of (GM.Gen.Xm u : GM.Gen c) : GM.G c) * (PresentedGroup.of (GM.Gen.Xm v : GM.Gen c) :
  ↪ GM.G c)
  = (PresentedGroup.of (GM.Gen.Xm (u + v) : GM.Gen c) : GM.G c) := by
  have h := PresentedGroup.mk_eq_mk_of_mul_inv_mem (GM.R1.re1a (c := c) u v)
  simp [GM.fXm, GM.G, PresentedGroup.of] using h
```

Proof. Put $a = X_{-1}(u)X_{-1}(v)$ and $b = X_{-1}(u + v)$ in $F(\mathcal{X}_c)$. By (Re:1a), the word ab^{-1} belongs to $\mathcal{R}_c$, hence to $N_{\mathcal{R}_c}$. Two words whose quotient ab^{-1} lies in $N_{\mathcal{R}_c}$ have the same image under π , so $\pi(a) = \pi(b)$. Since π is a homomorphism, $\pi(a) = \pi(X_{-1}(u))\pi(X_{-1}(v))$, which is the claim. $\square$

Lemma 3.3 (`Xm_zero_in_G`). *Let $c : \mathbb{N} \rightarrow \mathbb{N}$. Then $\pi(X_{-1}(0)) = 1$ in G_c .*

Lean: `Xm_zero_in_G` — proved by AIProver, job 2, session 1

```
theorem Xm_zero_in_G (c : ℕ → ℕ) :
  (PresentedGroup.of (GM.Gen.Xm (0 : ℤ) : GM.Gen c) : GM.G c) = 1 := by
  have h := PresentedGroup.one_of_mem (GM.Rl.re1a (c := c) (0 : ℤ) (0 : ℤ))
  simp [add_zero] using h
```

Proof. The instance $u = v = 0$ of (Re:1a) is the relator $X_{-1}(0)X_{-1}(0)X_{-1}(0 + 0)^{-1} \in \mathcal{R}_c$, so its image under π is 1. Since $0 + 0 = 0$, this word equals $X_{-1}(0)X_{-1}(0)X_{-1}(0)^{-1} = X_{-1}(0)$ in $F(\mathcal{X}_c)$. Hence $\pi(X_{-1}(0)) = 1$. $\square$

3.2 Proof of the theorem

Lean: `solution` — main proof from the captain's sketch

```
theorem solution (c : ℕ → ℕ) (u : ℤ) :
  ((PresentedGroup.of (GM.Gen.Xm (c := c) u) : GM.G c))-1 =
  (PresentedGroup.of (GM.Gen.Xm (c := c) (-u)) : GM.G c) := by
  apply inv_eq_of_mul_eq_one_right
  have h := Xm_add_in_G c u (-u)
  rw [add_neg_cancel, Xm_zero_in_G] at h
  exact h
```

Theorem 3.4. *For every $c : \mathbb{N} \rightarrow \mathbb{N}$ and every $u \in \mathbb{C}$, $\pi(X_{-1}(u))^{-1} = \pi(X_{-1}(-u))$ in G_c .*

Proof. In any group, if $xy = 1$ then $x^{-1} = y$. So it suffices to show that $\pi(X_{-1}(u))\pi(X_{-1}(-u)) = 1$. By Lemma 3.2 with $v = -u$, then $u + (-u) = 0$, then Lemma 3.3,

$$\pi(X_{-1}(u))\pi(X_{-1}(-u)) = \pi(X_{-1}(u + (-u))) = \pi(X_{-1}(0)) = 1.$$

$\square$

4 Independent review

A referee model (claude-opus-5-5), which took no part in the formalization or the proof, compared the verified Lean file with the source. Verdict: **FAITHFUL**.

Definitions. I checked every Lean definition against the source setting.

- `Idx` encodes E as triples (ℓ, j, k) with $\ell < j$ and $1 \leq k \leq c(j)$. Here $c : \mathbb{N} \rightarrow \mathbb{N}$ is a parameter, which the formalization note permits. If $j = 0$ there is no admissible ℓ , so including 0 in $\mathbb{N}$ adds nothing.
- `Gen` is the symbol set $\mathcal{X}$. H_1, H_2 are indexed by units of $\mathbb{C}$, and the unipotent symbols by $u \in \mathbb{C}$ and $e \in E$.
- `cc` is $c_{\ell j} = (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell)$. The natural-number subtractions $j-1$ and $j-\ell$ are exact because $\ell < j$. The value is nonzero, so the divisions by `cc` are genuine.
- `wm` and `wt` match $(\text{Re}:0)$ and $(\text{Im}:0)$.
- `comm` is $aba^{-1}b^{-1}$, as in the source, and `conj` is aba^{-1} .
- Each constructor of `R1` encodes $\text{lhs} = \text{rhs}$ as $\text{lhs} \cdot \text{rhs}^{-1}$. I compared them one by one with (H:1a) through (U:4b). Signs, exponents and the integer-power casts in (Im:5), (Im:6c) and (Im:6d) are correct. Side conditions are correct: $\ell+1=j$ for (U:1a,b); $\ell=0$ for (U:1c,d); the disjunction for (U:2); the indices $(1,2,k), (0,2,k)$ for (U:3a,b); the index $(j-1-\ell, j, k)$ with sign $(-1)^{j-\ell-1}$ for (U:4a,b).
- `G` is `PresentedGroup` on these relators, i.e. $F(\mathcal{X})/N_{\mathcal{R}}$.

I found no discrepancy.

Statement. The target states that for every $c : \mathbb{N} \rightarrow \mathbb{N}$ and every $u \in \mathbb{C}$, $X_{-1}(u)^{-1} = X_{-1}(-u)$ in $G(c)$.

- The source fixes c to be the coefficients of J . Instantiating the Lean theorem at that c recovers the source statement exactly, so nothing extra must be supplied.
- No hypothesis such as $c(2) \geq 1$ is needed here, and none is assumed.
- The conclusion is an equality in the presented group. It is not vacuous: it is exactly the corollary's claim.

Proof. The proof has three auxiliary lemmas and a final step.

- `mk_rel_eq_one_in_G` is unused.
- `Xm_add_in_G` derives $X_{-1}(u)X_{-1}(v) = X_{-1}(u+v)$ in G from relator (Re:1a). This is the first step of the source proof.
- `Xm_zero_in_G` derives $X_{-1}(0) = 1$ from (Re:1a) at $u = v = 0$, via $X(0)X(0)X(0)^{-1} = 1$. This plays the role of the cited Lemma 3.1.
- `solution` sets $v = -u$ to get $X_{-1}(u)X_{-1}(-u) = 1$, then concludes $X_{-1}(u)^{-1} = X_{-1}(-u)$ by uniqueness of inverses.

This follows the source proof directly, and each lemma states exactly the step it represents.

Concerns. None

A Complete Lean file

The verified file, as checked by Lean.

```
import Mathlib
set_option autoImplicit false

namespace GM

noncomputable section

/-- Index set  $E = \{(\ell, j, k) \mid 1 \leq k \leq c(j), 0 \leq \ell < j\}$ , encoded as triples  $(\ell, j, k)$ . -/
def Idx (c :  $\mathbb{N} \rightarrow \mathbb{N}$ ) : Type := {t :  $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$  // t.1 < t.2.1  $\wedge$  1  $\leq$  t.2.2  $\wedge$  t.2.2  $\leq$  c t.2.1}

variable {c :  $\mathbb{N} \rightarrow \mathbb{N}$ }

def Idx.l (e : Idx c) :  $\mathbb{N}$  := e.1.1
def Idx.j (e : Idx c) :  $\mathbb{N}$  := e.1.2.1
def Idx.k (e : Idx c) :  $\mathbb{N}$  := e.1.2.2

/-- The generating symbols  $\mathcal{X}$ . -/
inductive Gen (c :  $\mathbb{N} \rightarrow \mathbb{N}$ ) : Type
| H1 (s :  $\mathbb{C}^\times$ ) : Gen c
| H2 (s :  $\mathbb{C}^\times$ ) : Gen c
| Xm (u :  $\mathbb{C}$ ) : Gen c
| Ym (u :  $\mathbb{C}$ ) : Gen c
| X (e : Idx c) (u :  $\mathbb{C}$ ) : Gen c
| Y (e : Idx c) (u :  $\mathbb{C}$ ) : Gen c

def fH1 (s :  $\mathbb{C}^\times$ ) : FreeGroup (Gen c) := FreeGroup.of (Gen.H1 s)
def fH2 (s :  $\mathbb{C}^\times$ ) : FreeGroup (Gen c) := FreeGroup.of (Gen.H2 s)
def fXm (u :  $\mathbb{C}$ ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Xm u)
def fYm (u :  $\mathbb{C}$ ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Ym u)
def fX (e : Idx c) (u :  $\mathbb{C}$ ) : FreeGroup (Gen c) := FreeGroup.of (Gen.X e u)
def fY (e : Idx c) (u :  $\mathbb{C}$ ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Y e u)

/--  $c_{\{\ell\ j\}} = (-1)^{\{\ell+1\}} C(j-1, \ell) (\ell+1)(j-\ell)$ . -/
def cc (e : Idx c) :  $\mathbb{C}$  :=
  ((-1 :  $\mathbb{C}$ ) ^ (e.1 + 1)) * ((e.j - 1).choose e.1 :  $\mathbb{N}$ ) :  $\mathbb{C}$  * ((e.1 :  $\mathbb{C}$ ) + 1) * ((e.j - e.1 :  $\mathbb{N}$ ) :  $\mathbb{C}$ )

/-- (Re:0) -/
def wm (s :  $\mathbb{C}^\times$ ) : FreeGroup (Gen c) := fXm (s :  $\mathbb{C}$ ) * fYm (-(s :  $\mathbb{C}$ )-1) * fXm (s :  $\mathbb{C}$ )

/-- (Im:0) -/
def wt (e : Idx c) (s :  $\mathbb{C}^\times$ ) : FreeGroup (Gen c) :=
  fX e (s :  $\mathbb{C}$ ) * fY e (-(s :  $\mathbb{C}$ )-1) / cc e * fX e (s :  $\mathbb{C}$ )

/-- commutator (a,b) = a b a-1 b-1 -/
def comm (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a-1 * b-1

/-- conjugation a b a-1 -/
def conj (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a-1

/-- The set of defining relators; each constructor says `lhs = rhs` via the word `lhs * rhs-1`. -/
inductive R1 (c :  $\mathbb{N} \rightarrow \mathbb{N}$ ) : FreeGroup (Gen c) → Prop
| h1a (s t :  $\mathbb{C}^\times$ ) : R1 c (fH1 s * fH1 t * (fH1 (s * t))-1)
```

```

| h1b (s t :  $\mathbb{C}^\times$ ) : R1 c (fH2 s * fH2 t * (fH2 (s * t))-1)
| h2 (s t :  $\mathbb{C}^\times$ ) : R1 c (fH1 s * fH2 t * (fH2 t * fH1 s)-1)
| re1a (u v :  $\mathbb{C}$ ) : R1 c (fXm u * fXm v * (fXm (u + v))-1)
| re1b (u v :  $\mathbb{C}$ ) : R1 c (fYm u * fYm v * (fYm (u + v))-1)
| re2 (s t :  $\mathbb{C}^\times$ ) :
  R1 c (fYm (-(t :  $\mathbb{C}$ )) * fXm (s :  $\mathbb{C}$ ) * fYm (t :  $\mathbb{C}$ ) *
    (fXm (-(t :  $\mathbb{C}$ )-1) * fYm (-(t :  $\mathbb{C}$ ) ^ 2 * (s :  $\mathbb{C}$ )) * fXm ((t :  $\mathbb{C}$ )-1)-1)
| re3 (s :  $\mathbb{C}^\times$ ) : R1 c (wm s * wm 1 * (fH1 (-s) * fH2 (-s)-1)-1)
| re4a (u :  $\mathbb{C}$ ) : R1 c (conj (wm 1) (fXm u) * (fYm (-u))-1)
| re4b (u :  $\mathbb{C}$ ) : R1 c (conj (wm 1) (fYm u) * (fXm (-u))-1)
| re5a (s :  $\mathbb{C}^\times$ ) : R1 c (conj (wm 1) (fH1 s) * (fH2 s)-1)
| re5b (s :  $\mathbb{C}^\times$ ) : R1 c (conj (wm 1) (fH2 s) * (fH1 s)-1)
| re6a (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) : R1 c (conj (fH1 s) (fXm u) * (fXm ((s :  $\mathbb{C}$ ) * u))-1)
| re6b (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) : R1 c (conj (fH2 s) (fXm u) * (fXm ((s :  $\mathbb{C}$ )-1 * u))-1)
| re6c (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) : R1 c (conj (fH1 s) (fYm u) * (fYm ((s :  $\mathbb{C}$ )-1 * u))-1)
| re6d (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) : R1 c (conj (fH2 s) (fYm u) * (fYm ((s :  $\mathbb{C}$ ) * u))-1)
| im1a (e : Idx c) (u v :  $\mathbb{C}$ ) : R1 c (fX e u * fX e v * (fX e (u + v))-1)
| im1b (e : Idx c) (u v :  $\mathbb{C}$ ) : R1 c (fY e u * fY e v * (fY e (u + v))-1)
| im2 (e : Idx c) (s t :  $\mathbb{C}^\times$ ) :
  R1 c (fY e (-(t :  $\mathbb{C}$ )) * fX e (s :  $\mathbb{C}$ ) * fY e (t :  $\mathbb{C}$ ) *
    (fX e (-(t :  $\mathbb{C}$ )-1) / cc e) * fY e (-(cc e) * (t :  $\mathbb{C}$ ) ^ 2 * (s :  $\mathbb{C}$ )) *
    fX e ((t :  $\mathbb{C}$ )-1 / cc e)-1)
| im3 (e : Idx c) (s :  $\mathbb{C}^\times$ ) :
  R1 c (wt e (s ^ ((e.l + 1) * (e.j - e.l))) * wt e 1 *
    (fH1 ((-s) ^ (e.j - e.l)) * fH2 ((-s) ^ (e.l + 1)))-1)
| im4a (e : Idx c) (u :  $\mathbb{C}$ ) : R1 c (conj (wt e 1) (fX e u) * (fY e (-u / cc e))-1)
| im4b (e : Idx c) (u :  $\mathbb{C}$ ) : R1 c (conj (wt e 1) (fY e u) * (fX e (-(cc e) * u))-1)
| im5a (e : Idx c) (s :  $\mathbb{C}^\times$ ) :
  R1 c (conj (wt e 1) (fH1 (s ^ (e.j - e.l))) * (fH2 (s ^ (-(e.l :  $\mathbb{Z}$ ) + 1))))-1)
| im5b (e : Idx c) (s :  $\mathbb{C}^\times$ ) :
  R1 c (conj (wt e 1) (fH2 (s ^ (e.l + 1))) * (fH1 (s ^ (-(e.j :  $\mathbb{Z}$ ) - e.l))))-1)
| im6a (e : Idx c) (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) :
  R1 c (conj (fH1 s) (fX e u) * (fX e ((s :  $\mathbb{C}$ ) ^ (e.l + 1) * u))-1)
| im6b (e : Idx c) (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) :
  R1 c (conj (fH2 s) (fX e u) * (fX e ((s :  $\mathbb{C}$ ) ^ (e.j - e.l) * u))-1)
| im6c (e : Idx c) (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) :
  R1 c (conj (fH1 s) (fY e u) * (fY e ((s :  $\mathbb{C}$ ) ^ (-(e.l :  $\mathbb{Z}$ ) + 1) * u))-1)
| im6d (e : Idx c) (s :  $\mathbb{C}^\times$ ) (u :  $\mathbb{C}$ ) :
  R1 c (conj (fH2 s) (fY e u) * (fY e ((s :  $\mathbb{C}$ ) ^ (-(e.j :  $\mathbb{Z}$ ) - e.l) * u))-1)
| u1a (e : Idx c) (h : e.l + 1 = e.j) (u v :  $\mathbb{C}$ ) : R1 c (comm (fXm u) (fX e v))
| u1b (e : Idx c) (h : e.l + 1 = e.j) (u v :  $\mathbb{C}$ ) : R1 c (comm (fYm u) (fY e v))
| u1c (e : Idx c) (h : e.l = 0) (u v :  $\mathbb{C}$ ) : R1 c (comm (fYm u) (fX e v))
| u1d (e : Idx c) (h : e.l = 0) (u v :  $\mathbb{C}$ ) : R1 c (comm (fXm u) (fY e v))
| u2 (e f : Idx c) (h : e.j  $\neq$  f.j  $\vee$  e.k  $\neq$  f.k  $\vee$  1 < ((e.l :  $\mathbb{Z}$ ) - f.l)).natAbs) (u v :  $\mathbb{C}$ ) :
  R1 c (comm (fX e u) (fY f v))
| u3a (e f : Idx c) (k :  $\mathbb{N}$ ) (he : e.l = (1, 2, k)) (hf : f.l = (0, 2, k)) (u v :  $\mathbb{C}$ ) :
  R1 c (comm (fX e u) (fY f v) * (fXm (u * v))-1)
| u3b (e f : Idx c) (k :  $\mathbb{N}$ ) (he : e.l = (0, 2, k)) (hf : f.l = (1, 2, k)) (u v :  $\mathbb{C}$ ) :
  R1 c (comm (fX e u) (fY f v) * (fYm (u * v))-1)
| u4a (e e' : Idx c) (h : e'.1 = (e.j - 1 - e.l, e.j, e.k)) (u :  $\mathbb{C}$ ) :
  R1 c (conj (wm 1) (fX e u) * (fX e' ((-1 :  $\mathbb{C}$ ) ^ (e.j - e.l - 1) * u))-1)
| u4b (e e' : Idx c) (h : e'.1 = (e.j - 1 - e.l, e.j, e.k)) (u :  $\mathbb{C}$ ) :
  R1 c (conj (wm 1) (fY e u) * (fY e' ((-1 :  $\mathbb{C}$ ) ^ (e.j - e.l - 1) * u))-1)

```

-- The group $G(m) = \langle \mathcal{X} \mid \mathcal{R} \rangle$. -/

abbrev G (c : $\mathbb{N} \rightarrow \mathbb{N}$) : **Type** := PresentedGroup {r : FreeGroup (Gen c) | R1 c r}

end

end GM

theorem mk_rel_eq_one_in_G (c : $\mathbb{N} \rightarrow \mathbb{N}$) (r : FreeGroup (GM.Gen c)) (h : GM.R1 c r) :
 (PresentedGroup.mk {r' : FreeGroup (GM.Gen c) | GM.R1 c r'} r : GM.G c) = 1 := **by**
 exact (PresentedGroup.one_of_mem h)

theorem Xm_add_in_G (c : $\mathbb{N} \rightarrow \mathbb{N}$) (u v : $\mathbb{C}$) :
 (PresentedGroup.of (GM.Gen.Xm u : GM.Gen c) : GM.G c) * (PresentedGroup.of (GM.Gen.Xm v : GM.Gen c) :
 $\hookrightarrow$ GM.G c)
 = (PresentedGroup.of (GM.Gen.Xm (u + v) : GM.Gen c) : GM.G c) := **by**
 have h := PresentedGroup.mk_eq_mk_of_mul_inv_mem (GM.R1.re1a (c := c) u v)
 simp [GM.fXm, GM.G, PresentedGroup.of] **using** h

theorem Xm_zero_in_G (c : $\mathbb{N} \rightarrow \mathbb{N}$) :
 (PresentedGroup.of (GM.Gen.Xm (0 : $\mathbb{C}$) : GM.Gen c) : GM.G c) = 1 := **by**
 have h := PresentedGroup.one_of_mem (GM.R1.re1a (c := c) (0 : $\mathbb{C}$) (0 : $\mathbb{C}$))
 simp [add_zero] **using** h

theorem solution (c : $\mathbb{N} \rightarrow \mathbb{N}$) (u : $\mathbb{C}$) :
 ((PresentedGroup.of (GM.Gen.Xm (c := c) u) : GM.G c))⁻¹ =
 (PresentedGroup.of (GM.Gen.Xm (c := c) (-u)) : GM.G c) := **by**
 apply inv_eq_of_mul_eq_one_right
 have h := Xm_add_in_G c u (-u)
 rw [add_neg_cancel, Xm_zero_in_G] **at** h
 exact h